🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 10 · STANDARD 6.GR.2

Volume with Whole Number Edges

Lesson 10-1 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can find the volume of a rectangular prism with whole-number edges using length × width × height.
Language Objective: I can explain my work using the words volume, rectangular prism, cubic units, and edge.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Volume | Rectangular prism | Cubic units | Length, width, height | Net | Edge

👆 *Tap any word to see what it means and a picture.*

Level 3: try the blanks from memory first, then check the bank.

1. The amount of space inside a three-dimensional solid is its ________.

2. A solid with six rectangular faces, like a box, is a ________.

3. The unit used to measure volume, like cubic centimeters, is ________.

4. The three edge measurements of a rectangular prism are its ________.

5. A flat pattern that folds up into a solid is a ________.

6. A line segment where two faces of a solid meet is an ______________________.

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

Container C is 8 × 3 × 4 and Container B is 5 × 5 × 2. How did you count or multiply the unit cubes to find each volume, and which holds more?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Volume** — How much space is inside a solid shape.
  Volumen — Cuánto espacio hay dentro de una figura sólida.
- [ ] **Rectangular prism** — A solid box shape with six flat rectangle sides.
  Prisma rectangular — Una figura sólida en forma de caja con seis lados rectangulares.
- [ ] **Cubic units** — The units used to measure space inside, like cubic inches.
  Unidades cúbicas — Las unidades para medir el espacio interior, como pulgadas cúbicas.
- [ ] **Length, width, height** — How long, how wide, and how tall a box is.
  Largo, ancho, altura — Qué tan largo, ancho y alto es una caja.
- [ ] **Net** — A flat shape that folds up into a solid.
  Plantilla (desarrollo plano) — Una figura plana que se dobla y forma un sólido.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

I found the volume by multiplying ____ × ____ × ____, which equals ____ cubic units.
Hallé el volumen multiplicando ____ × ____ × ____, que es ____ unidades cúbicas.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: Volume measures space in three directions, so you need length, width, and height.

Evidence: Students explain that volume is three-dimensional, so all three edges (length, width, height) are needed; two measurements only give area of one face.

Reasoning: This evidence connects the definition of volume to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **I found the volume by multiplying ____ × ____ × ____, which equals ____ cubic units.**
  Hallé el volumen multiplicando ____ × ____ × ____, que es ____ unidades cúbicas.
- **Container ____ holds more because its volume is ____.**
  El contenedor ____ contiene más porque su volumen es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Volume
2. **Notes 2:** Rectangular prism
3. **Notes 3:** Cubic units
4. **Notes 4:** Length, width, height
5. **Notes 5:** Net
6. **Notes 6:** Edge
7. **Watch for this mistake:** *A common mistake in Volume with Whole Number Edges is multiplying only two of the three edges and stopping there — for example, seeing a box that is 6 in long, 4 in wide, and 3 in tall, multiplying just 6 × 4 = 24, and calling that the volume, when 24 is only the area of the bottom face (in square inches), not the volume of the whole box. The height must also be multiplied in: V = 6 × 4 × 3 = 72 cubic inches. A second common mistake is adding the three edges instead of multiplying them (6 + 4 + 3 = 13 instead of 6 × 4 × 3 = 72 cubic inches). Volume always MULTIPLIES all three edges, and the answer is always labeled in cubic units (in^3), never square units (in^2) or plain units (in).*
8. **Try It 1:** A. 30 cubic units — *V = length × width × height = 5 × 3 × 2 = 30 cubic units.*
9. **Try It 2:** A. 24 cubic units — *Each layer has 6 cubes and there are 4 layers, so 6 × 4 = 24 cubic units. This is the same as length × width × height with one factor being the number of layers.*